frequency domain is complex and periodic, the appropriate section is over one complete cycle, i.e., $-\pi$ to π (continuous frequency domain), or 0 to N-1 (discrete frequency domain). If the frequency domain is real and aperiodic, the appropriate section is zero to positive infinity, that is, only the positive frequencies. Lastly, if the frequency domain is real and periodic, the appropriate section is over the one-half cycle containing the positive frequencies, either 0 to π (continuous frequency domain) or 0 to $N/2$ (discrete frequency domain).

8. Scaling

To make the analysis and synthesis equations undo each other, a scaling factor must be placed on one or the other equation. In Table 31-1, we have placed the scaling factors with the analysis equations. In the complex case, these scaling factors are: $1/N$, $1/T$, or $1/2\pi$. Since the real transforms do not use negative frequencies, the scaling factors are twice as large: $2/N$, $2/T$, or $1/\pi$. The real transforms also include a negative sign in the calculation of the imaginary part of the frequency spectrum (an option used to make the real transforms more consistent with the complex transforms). Lastly, the synthesis equations for the real DFT and the real Fourier Series have special scaling instructions involving $Re\,X(0)$ and $Re\,X[N/2]$.

9. Variations

These equations may look different in other publications. Here are a few variations to watch out for:

- Using f instead of ω by the relation: $\omega = 2\pi f$
- Integrating over other periods, such as: $-T$ to 0, $-T/2$ to $T/2$, or 0 to T
- Moving all or part of the scaling factor to the synthesis equation
- Replacing the period with the fundamental frequency, $f_0 = 1/T$
- Using other variable names, for example, ω can become Ω in the DTFT, and $Re\,X[k]$ & $Im\,X[k]$ can become a_k & b_k in the Fourier Series

Why the Complex Fourier Transform is Used

It is painfully obvious from this chapter that the complex DFT is much more complicated than the real DFT. Are the benefits of the complex DFT really worth the effort to learn the intricate mathematics? The answer to this question depends on who you are, and what you plan on using DSP for. A basic premise of this book is that most practical DSP techniques can be understood and used without resorting to complex transforms. If you are learning DSP to assist in your non-DSP research or engineering, the complex DFT is probably overkill.

Nevertheless, complex mathematics is the primary language of those that specialize in DSP. If you do not understand this language, you cannot communicate with professionals in the field. This includes the ability to understand the DSP literature: books, papers, technical articles, etc. Why are complex techniques so popular with the professional DSP crowd?